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% Input
mu = 0.3; % Average friction coefficient
C = 0; % Cohesion (MPa), Pre-faulted plane

% Dip angle range
theta_deg = linspace(1, 35, 100); % max dip angle on the MMT
theta = deg2rad(theta_deg);

% Stress states: sigma1 = Sh (max, horizontal), sigma3 = Sv (min, vertical)

depth_km = [1, 2];
Sv = [25, 55]; % sigma3 (MPa)
Sh = [55, 100]; % sigma1 (MPa)

% Pore pressure scenarios
Pf_hydro = [9.8, 20]; % hydrostatic pressure (MPa), ~1 km & 2 Km
Pf_litho = Sv; % lithostatic pressure (MPa) Pf = Sv

% figure
figure('Position',);

for i = 1:numel(depth_km)

    sigma1 = Sh(i);
    sigma3 = Sv(i);

    % Normal and shear stress resolved on the fault plane
    sigma_n = sigma1 .* (sin(theta)).^2 + sigma3 .* (cos(theta)).^2;
    tau = (sigma1 - sigma3) .* sin(theta) .* cos(theta);

    % Coulomb failure envelope
    tau_c = mu .* sigma_n + C;

    % Coulomb envelope @ hydrostatic pore pressure
    sigma_n_eff_hydro = sigma_n - Pf_hydro(i);
    tau_c_eff_hydro = mu .* sigma_n_eff_hydro + C;

    % Coulomb envelope @ lithostatic pore pressure
    sigma_n_eff_litho = sigma_n - Pf_litho(i);
    tau_c_eff_litho = mu .* sigma_n_eff_litho + C;

    subplot(1, 2, i)
    plot(theta_deg, sigma_n, 'b-', 'LineWidth', 2, 'DisplayName', '\sigma_n normal stress')
    hold on
    plot(theta_deg, tau, '--', 'Color', [0.8500 0.3250 0.0980], 'LineWidth', 2, ...
        'DisplayName', sprintf('\tau resolved (\sigma_1=%d, \sigma_3=%d MPa)', sigma1, sigma3))
    plot(theta_deg, tau_c, 'k:', 'LineWidth', 2, ...
        'DisplayName', '\tau Coulomb = \mu\sigma_n + C')
    plot(theta_deg, tau_c_eff_hydro, 'g--', 'LineWidth', 2, ...
        'DisplayName', sprintf('\tau Coulomb, P_f=%0.1f MPa (hydrostatic)', Pf_hydro(i)))
    plot(theta_deg, tau_c_eff_litho, 'm-', 'LineWidth', 2, ...
        'DisplayName', sprintf('\tau Coulomb, P_f=%0.1f MPa (lithostatic)', Pf_litho(i)))
    hold off

    xlabel('Fault dip angle (degrees)', 'FontSize', 12)
    ylabel('Stress (MPa)', 'FontSize', 12)

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title(sprintf('%d km depth (\\sigma_1=S_h=%d MPa, \\sigma_3=S_v=%d MPa, \\mu=%.1f, C=%.1f MPa)', ...  
    depth_km(i), sigma1, sigma3, mu, C), 'FontSize', 12)  
legend('Location', 'best', 'FontSize', 10)  
grid on  
end
```